

Week		Date	Day	Topic & Skill	Content Coverage
1	1	17 August 2026	Mon	Programming Fundamentals + Database & RDBMS	Planning Logic – Pseudocode, flowcharts Programming Basics – What is programming, types of languages Syntax & Structure – Variables, data types, operators Control Structures – if, else, loops (for, while) Functions and Modular Programming – Defining, calling, parameters, return values Database: Introduction to Databases What is a Database? Types of Databases: Relational vs Non-Relational Overview of RDBMS (Relational Database Management System) Popular RDBMS Tools: MySQL, PostgreSQL, Oracle, SQL Server, Teradata SQL Basics SQL Syntax and Structure Data Definition Language (DDL): CREATE, ALTER, DROP, TRUNCATE
	2	18 August 2026	Tue	Database & RDBMS	Database contd.: Data Manipulation Language (DML): INSERT, UPDATE, DELETE Data Query Language (DQL): SELECT, WHERE, FROM, ORDER BY, GROUP BY Advanced SQL Joins: Inner, Left, Right, Full [Focus] Subqueries and Nested Queries Transactions and Concurrency ACID Properties Transaction Control: COMMIT, ROLLBACK, SAVEPOINT [HoL must] Database Administration - server backups User Roles and Permissions
	3	19 August 2026	Wed	SQL Fundamentals & Intermediate SQL	Operators: AND, OR, NOT, BETWEEN, IN, LIKE Recap Joins: INNER, LEFT, RIGHT, FULL OUTER, SELF JOIN Recap Subqueries and Nested Queries [Focus] Aliases and Set Operations (UNION, INTERSECT, MINUS) Aggregate Functions: COUNT, SUM, AVG, MIN, MAX Views and Indexing Data Control Language (DCL): GRANT, REVOKE DQL - HAVING, LIMIT
	4	20 August 2026	Thu	Advanced SQL Techniques	Advanced SQL Techniques Window Functions: ROW_NUMBER(), RANK(), LEAD(), LAG() Case Expressions and Conditional Logic Stored Procedures and Triggers
	5	21 August 2026	Fri	SQL Practical Applications	Practical Applications Real-world Projects: Inventory System/Analytics Dashboard SQL for Data Analytics and Reporting Working with Large Datasets
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2	6	24 August 2026	Mon	Advanced Analytical Topics & Recursive CTE + Weekly Assessment-1	Advanced Analytical Topics Query Optimization and Execution Plans Temporary Tables and Clustering Common Table Expressions (CTEs) Recursive CTEs Recursive CTE for Organizational Hierarchies
	7	25 August 2026	Tue	Performance Benchmarking & DWH Concepts	Performance Benchmarking Data Warehouse (DWH) Concepts Introduction to Data Warehousing What is a Data Warehouse? OLTP vs OLAP systems Benefits and use cases of DWH Key components of a DWH system Introduction to snowflake Data warehouse lifecycle [SQL on snowflake labs needed - single database]
	8	26 August 2026	Wed	DWH Architecture & data modelling	Data Warehouse Architecture Basic architecture: staging, integration, access layers Enterprise Data Warehouse (EDW) vs Data Marts ETL vs ELT processes Real-time vs batch data warehousing [batch data lab needed] Cloud-based DWH (e.g., Snowflake, BigQuery, Redshift, Databricks) Data Modeling Dimensional modeling: Star and Snowflake schemas [Lab needed] Fact and Dimension tables Slowly Changing Dimensions (SCD Types 1, 2, 3) Granularity and surrogate keys Normalization vs denormalization SQL on snowflake labs needed - multiple databases (fact/dimension tables implementation)
	9	27 August 2026	Thu	ETL/ELT & data profiling	ETL/ELT Processes Overview of ETL tools (Informatica, Qlik Talend, Fivetran, Matillion etc.) Data extraction from source systems Data transformation techniques Data loading strategies Error handling and logging Data Integration, Quality and BI concepts Data profiling and cleansing Handling duplicates and missing data [Compute management lab needed] [Query + visualization lab needed]

	10	28 August 2026	Fri	Cloud Data Warehousing	Cloud Data Warehousing Introduction to cloud DWH platforms Snowflake, Databricks and Google BigQuery and Redshift overview Scalability Sample Real-World Projects
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3	11	31 August 2026	Mon	Mocktest-1 and Debriefing	
	12	01 September 2026	Tue	Python Fundamentals	Introduction to Python and its ecosystem Installing Python and setting up IDEs (VS Code, PyCharm, Jupyter) Basic syntax, variables, and data types - int, str, float, var, list Operators and expressions Input/output operations Comments and code documentation
	13	02 September 2026	Wed	Control Structures & Loops	Data Structures Lists, Tuples Control Structures Conditional statements (if, elif, else) Loops (for, while) Loop control (break, continue, pass) Comprehensions (list, dict, set) Regular Expressions
	14	03 September 2026	Thu	Functions and Modules	Functions and Modules Defining and calling functions Arguments and return values Lambda functions Recursion Modules and packages Importing and using libraries
	15	04 September 2026	Fri	Data Structures	Data Structures Recap - Lists, Tuples Set, Dictionaries Operations and methods Nested structures Iteration and manipulation
			Sat		
			Sun		

4	16	07 September 2026	Mon	Object-Oriented Programming (OOP) Working with libraries	Object-Oriented Programming (OOP) Classes and objects Constructors and destructors Inheritance and polymorphism Encapsulation and abstraction Magic methods (__init__, __str__, etc.) Working with Libraries NumPy for numerical data analysis [numpy focus]
	17	08 September 2026	Tue	Numpy & Pandas	NumPy and Pandas for data manipulation [pandas focus]
	18	09 September 2026	Wed	Weekly Assessment-2 + Working with libraries - Matplotlib, Seaborn, SQLite/SQLAlchemy	Matplotlib and Seaborn for visualization SQLite/SQLAlchemy for database interaction [1 SQL connectivity lab needed]
	19	10 September 2026	Thu	Exception & File Handling	Exception Handling Try, Except, Finally blocks Custom exceptions Assertions Logging and debugging File Handling Reading and writing files Working with CSV, JSON, XML File modes and context managers Directory and file operations
	20	11 September 2026	Fri	Testing & deployment + Data Analysis Project	Testing and Deployment Unit testing with unit test and pytest Data analysis project (e.g., COVID trends) Automation scripts (e.g., email sender, file organizer)
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		14 September 2026	Mon	Ganesh Chaturthi (Holiday)	
5	21	15 September 2026	Tue	Data Analysis Project - Contd.	Data analysis project (e.g., COVID trends) Automation scripts (e.g., email sender, file organizer)
	22	16 September 2026	Wed	AI Essentials	Essential AI Topics Introduction to AI - What is AI, ML, DL, AGI Prompt Engineering & ChatGPT - What is prompt engineering - ChatGPT use cases (e.g., for data analysis) Building Generative AI Applications - Prototyping and deploying apps - Evaluation techniques for NLP models
	23	17 September 2026	Thu	Mocktest-2 and Debriefing	
	24	18 September 2026	Fri	Milestone 1 Assessment	
			Sat		

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6	25	21 September 2026	Mon	PowerBI Fundamentals - Getting Started and	1. Getting Started What is Power BI? Components: Desktop, Service, Mobile Installation & Licensing 2. Data Connectivity Connect to Excel, SQL, Azure, Web Import vs DirectQuery Latest connectors: Snowflake, BigQuery, Databricks Microsoft Fabric integration 3. Data Preparation Power Query Editor basics Data shaping: remove, unpivot, transform
	26	22 September 2026	Tue	PowerBI Fundamentals - Data Modelling and Visualizations	5. Data Modeling Relationships & schemas Hierarchies & calculated tables 4. Visualizations Charts, tables, slicers Formatting & publishing Custom visuals & bookmarks
	27	23 September 2026	Wed	PowerBI Fundamentals - Reports & Interactivity + Milestone 1 Retest (If any)	7. Reports & Interactivity Drillthrough, tooltips, sync slicers Performance tuning & query diagnostics
	28	24 September 2026	Thu	PowerBI Fundamentals - DAX (Data Analysis Expressions)	6. DAX (Data Analysis Expressions) Measures vs columns
	29	25 September 2026	Fri	PowerBI Fundamentals - DAX (Data Analysis Expressions)	6. DAX (Data Analysis Expressions) contd., Key functions: CALCULATE, FILTER, TIME intelligence Advanced DAX: variables, context, RANKX
			Sat		
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7	30	28 September 2026	Mon	PowerBI Advanced - Security + Weekly Assessment-3	8. Security Row-Level Security (RLS) Azure AD integration
	31	29 September 2026	Tue	PowerBI Advanced - Collaboration & Service	9. Collaboration & Service Workspaces, sharing, mobile access Embedded analytics
	32	30 September 2026	Wed	PowerBI Advanced -Microsoft Fabric & Real-Time	10. Microsoft Fabric & Real-Time OneLake, streaming datasets, Spark orchestration

	33	01 October 2026	Thu	PowerBI Advanced - Custom Development	11. Custom Development REST APIs, PBIP schemas Deployment pipelines Data refresh schedule
		02 October 2026	Fri	Gandhi Jayanthi (Holiday)	
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8	34	05 October 2026	Mon	PowerBI Advanced -Power BI Copilot and Natural language Q&A + PL300 certification (offline read material)	12. Power BI Copilot Setup & semantic models AI-generated visuals & DAX Natural language Q&A Governance & responsible AI PL300 certification (offline read material)
	35	06 October 2026	Tue	MS-2 Mocktest-1 and Debriefing	
	36	07 October 2026	Wed	Milestone 2 Assessment	
	37	08 October 2026	Thu	Tableau - Getting Started	1. Getting Started Tableau products: Desktop, Prep, Server Interface, data types, and connections (SQL, Snowflake, BigQuery, Excel, etc.)
	38	09 October 2026	Fri	Tableau - Visualizations & Dashboards	2. Visualizations & Dashboards Basic charts: bar, line, pie
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9	39	12 October 2026	Mon	Tableau - Visualizations & Dashboards + Milestone 2 Retest (If any)	2. Visualizations & Dashboards - contd. Filters, sorting, grouping Dashboards, stories, publishing
	40	13 October 2026	Tue	Tableau - Data Preparation	3. Data Preparation Tableau Prep basics Joins, unions, blending Pivoting, cleaning, shaping
	41	14 October 2026	Wed	Tableau - Calculated Fields & LOD	4. Calculated Fields & LOD Functions: numeric, string, conditional Table calculations LOD expressions: FIXED, INCLUDE, EXCLUDE
	42	15 October 2026	Thu	Tableau - Parameters & Advanced Visuals and Analytics & Optimization	5. Parameters & Advanced Visuals Dynamic filters, set actions What-if analysis Advanced charts: Sankey, radial, maps 6. Analytics & Optimization Forecasting, clustering, cohort analysis Extracts vs live connections Performance tuning

	43	16 October 2026	Fri	Tableau - Server & Governance and Tableau Agent & AI	7. Server & Governance Server setup, roles, permissions Publishing, scheduling, security 8. Tableau Agent & AI Agent: Enable, ask questions, get alerts Ask Data: Natural language queries, instant charts, share results
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10	44	19 October 2026	Mon	Qlik Analytics - Introduction and Data Loading & Preparation + Weekly Assessment-4	1. Introduction Qlik Sense vs QlikView Associative data model Setup and navigation 2. Data Loading & Preparation Load from Excel, databases, web Data Manager vs Load Editor Scripting basics: JOIN, KEEP, CONCATENATE
		20 October 2026	Tue	Vijya Dashmi(Holiday)	
	45	21 October 2026	Wed	Qlik Analytics - Visualizations	3. Visualizations Charts, tables, dashboards Storytelling and drill-downs
	46	22 October 2026	Thu	Qlik Analytics - Expressions & Set Analysis	4. Expressions & Set Analysis Aggregations, conditions Alternate states, advanced expressions
	47	23 October 2026	Fri	Qlik Analytics - Scripting & Variables	5. Scripting & Variables Advanced scripting Dynamic dimensions and modular code
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	48	26 October 2026	Mon	Qlik Analytics - Geo-Analytics and Dashboard Design	6. Geo-Analytics Maps, spatial functions, heatmaps 7. Dashboard Design CSS styling, responsive layouts UX and performance tips
11	49	27 October 2026	Tue	Qlik Analytics - Advanced Analytics	8. Advanced Analytics Predictive modeling Integration with R, Python, REST APIs

	50	28 October 2026	Wed	Qlik Analytics - Qlik Answers and AI assistants & usage	9.Qlik Answers a. Overview What is Qlik Answers? Use cases and AI technologies b. Setup Create knowledge bases Upload and index documents c. AI Assistants Link to knowledge bases Customize and embed assistants d. Usage Ask questions via chat Get explainable answers Improve responses over time
	51	29 October 2026	Thu	MS 3 Mocktest-1 and Debriefing	
	52	30 October 2026	Fri	Milestone 3 Assessment	
			Sat		
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	53	02 November 2026	Mon	Mock client Interview	
12	54	03 November 2026	Tue	Capstone Project	
	55	04 November 2026	Wed	Capstone Project	
	56	05 November 2026	Thu	Capstone Project	
	57	06 November 2026	Fri	Capstone Project	
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		09 November 2026	Monday	Diwali (Holiday)	
	58	10 November 2026	Tue	Capstone Project	
	59	11 November 2026	Wed	Milestone - 5 : Capstone Project Evaluation	
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				Capstone Project Re-Evaluation (If any)	